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CCNA Internship Report

Week 1 · Task 1 — Build Your CCNA Lab + How Networks Actually Work

1. Introduction

Static Routing is a networking method in which routes are manually configured on routers. It allows routers to forward data packets between different networks using predefined paths.

Cisco Packet Tracer is used to understand network communication, router configuration, IP addressing, and static routing.

2. Objectives

- To understand the concept of static routing.
- To learn how routers communicate with different networks.
- To understand IP addressing and subnet masks.
- To configure static routes using Cisco IOS commands.
- To verify network connectivity using the ping command.

3. Tools and Devices Used

- Cisco Packet Tracer
- 3 Routers
- 3 Switches
- 6 Laptops
- Network cables
- Router CLI

4. Network Topology

The network consists of three routers connected through WAN links. Each router connects to a separate LAN through a switch. Each LAN contains two end devices.

5. IP Addressing Scheme

The network uses the following IP addressing plan:

LAN Networks

- LAN 1: 192.168.1.0/24
- LAN 2: 192.168.2.0/24
- LAN 3: 192.168.3.0/24

WAN Links

- Router 1 to Router 2: 12.0.0.0
- Router 2 to Router 3: 23.0.0.0

6. Functions of Network Devices

Router: Connects different networks and forwards packets using routing information.

Switch: Connects multiple devices within the same local area network and forwards frames to the appropriate devices.

Laptop: Acts as an end device that sends and receives network data.

7. Router Configuration

Router interfaces use IP addresses to establish communication between connected networks.

Example interface configuration:

interface fastEthernet0/0

ip address 12.0.0.1 255.0.0.0

no shutdown

Static routes are added manually to inform routers about remote networks.

Router 1:

- ip route 23.0.0.0 255.0.0.0 12.0.0.2
- ip route 192.168.2.0 255.255.255.0 12.0.0.2
- ip route 192.168.3.0 255.255.255.0 12.0.0.2

Router 2:

- ip route 192.168.3.0 255.255.255.0 23.0.0.3
- ip route 192.168.1.0 255.255.255.0 12.0.0.1

Router 3:

- ip route 192.168.1.0 255.255.255.0 23.0.0.2
- ip route 12.0.0.0 255.0.0.0 23.0.0.2
- ip route 192.168.2.0 255.255.255.0 23.0.0.2

8. Connectivity Testing

The ping command is used to test communication between devices connected to different networks.

Command:

ping <destination-ip-address>

The routing table can also be checked using:

show ip route

Successful ICMP replies indicate that communication between the tested devices is working.

9. Packet Transmission Path

When a device sends data to another network, the packet is forwarded to its default gateway. The router checks its routing table and forwards the packet to the next-hop router.

The packet continues through the connected routers until it reaches the destination network. The destination switch then forwards the data to the required end device.

10. Observations

- Static routes provide manually defined paths between networks.
- Routers use routing tables to forward packets.
- Correct IP addresses and subnet masks are important for communication.
- The next-hop address helps routers identify where to forward packets.
- Ping is useful for checking end-to-end connectivity.

11. Conclusion

Static routing provides a simple way to connect multiple networks through manually configured routes. This lab demonstrates the role of routers, switches, IP addressing, routing tables, and ICMP testing in network communication. It also explains how data travels between different LANs using predefined routing paths

12.Screen Shots









